HEAT-TREAT REFERENCE — 10 STEELS

Working starting points for the ten steels makers actually use. Temperatures in °F. Temper twice, 2 hours per cycle, cool to room temperature between cycles.

STEEL	NORMALIZE	AUSTENITIZE	QUENCH	TEMPER	TARGET HRC	NOTES
1084	1600	1475–1500 · 10–15 min	Fast oil (Parks 50)	375–425	58–60	The beginner's benchmark. Forgiving, clean.
1095	1650	1475 · 10 min	FAST oil only	350–400	60–62	Needs speed — slow oil = soft spots. Thin edges warp.
80CrV2	1600	1500–1525 · 10–15 min	Fast-medium oil	400–450	57–59	Tough field-knife favorite. Takes a beating.
5160	1650	1525 · 10–15 min	Medium oil	400–450	57–58	Spring steel. Choppers, swords, big blades.
52100	1650 / 1600 / 1550	1500–1550 · short soak	Fast oil	350–400	60–62	Triple normalize pays off. Fine edge, ball-bearing steel.
W2	1600	1450–1475 · 10 min	Brine or fast oil	350–400	60–62	Shallow hardening — the hamon steel.
O1	1600	1450–1500 · 10–15 min	Medium oil	~400	60–61	Forgiving tool steel. Great in a basic shop.
D2	—	1850 · 30–45 min	Plate / still air	400–500	58–61	Foil wrap. Semi-stainless, carbide-rich, toothy edge.
AEB-L	—	1950 · 10–15 min	Plate quench	300–350	60–62	Foil wrap. Cryo or dry ice recommended. Razor steel.
MagnaCut	—	1950–2050 · 15–30 min	Plate quench	300–350	60.5–63.5	Foil + cryo strongly recommended. The modern do-it-all.

QUENCH MEDIA QUICK GUIDE

Parks 50 / fast oil	Shallow-hardening carbon steels: 1084, 1095, W2, 52100
Medium oil (AAA, canola @ 120°F)	O1, 5160, 80CrV2 — deeper hardening, slower is safer
Plate quench (aluminum + air)	Stainless + air-hardening: D2, AEB-L, MagnaCut
Brine	W2 hamons only — violent, crack risk, know why you're using it

HABITS THAT SAVE BLADES

Snap temper	Within an hour of the quench — 300–350°F, 30 min — before straightening or grinding.
Test coupon	Same steel, same batch, same cycle. File test + break test before you commit a blade.
Foil wrap	Everything over ~1800°F gets stainless foil or a coating — decarb eats edges.
Straighten warm	Carbon steel straightens between temper cycles at ~400°F, gloved hands, gentle.

Read this once: these are proven STARTING points, not gospel. Ovens lie by ±25°F, steel batches vary, and geometry changes everything. Verify against your steel supplier's datasheet and run a test coupon before trusting any number on this page with a finished blade.